Julian Henry

Houston, TX USA · github.com/juleshenry · juleshenry.github.io

Senior full-stack engineer building trading, telemetry, and security-analytics systems in C++, Python, Java, and TypeScript.

Work Experience

Senior Full-Stack Engineer — October 2024 – Present

Macquarie Financial Ltd. — *Houston, TX*

- Operate and harden Java servlet services that power critical trading workflows across London, São Paulo, and Sydney desks daily.
- Ported as-dealt C++ deal logic into Python services that power ASIC regulatory reporting and fully auditable end-to-end trade state.
- Co-author shared risk, booking, and workflow control frameworks spanning 100+ commodity and financial product lines firm-wide.
- Ship and harden production C++17 features across a 1M+ line platform engineered for low-latency, high-availability global trading.
- Partner directly with trading desks and compliance stakeholders to translate regulatory mandates into shippable, audit-ready systems.

Software Engineer — May 2023 – October 2024

AGCO Corporation — *Atlanta, GA*

- Hardened AWS and Kafka telemetry pipelines serving connected farm equipment fleets operating across 140 countries worldwide.
- Engineered RESTful APIs for hardware device orchestration, accelerating integration-test cycles across connected farm equipment.
- Shipped Angular features for the Fendt.One precision agriculture platform supporting 10,000+ remote farm operations worldwide.
- Aligned hardware, product, and QA teams around a single equipment-integration roadmap, cutting rework and speeding delivery.

Data Engineer — August 2020 – August 2022

SimSpace Corporation — *Boston, MA*

- Led adversarial red-team cyber-range campaigns to stress-test production ML threat-detection models across enterprise workloads.
- Built cybersecurity analytics pipelines with NLP that improved threat detection rates by 72% across enterprise security workloads.
- Optimized a 30,000+ line Python/Pandas ETL pipeline, delivering 10× throughput under production cybersecurity analytics loads.
- Prioritized which detection gaps red-team findings should drive next, steering the security team's ML roadmap investment decisions.

Assets & Liabilities Intern — June 2019 – May 2020

Transamerica Corporation — *Baltimore, MD · Part-time*

- Validated variable-annuity risk models in NumPy, scikit-learn, and pandas for actuarial stress testing and cashflow reconciliation.
- Achieved 99.7% simulation accuracy for reconciliation and stress testing across a $6B+ AUM variable annuity book of business.

Technical Skills

Languages: C++17, Python, Java, TypeScript, JavaScript, SQL
Frontend: React, Next.js, Angular, Tailwind CSS
Backend: Java Servlets, REST APIs, Node.js, Flask
AWS: Lambda, S3, EC2, RDS, CloudWatch
Cloud / Infra: Kafka, Docker, Terraform, Kubernetes, CI/CD
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch
Data / ML: Pandas, NumPy, scikit-learn, NLP, RAG, FAISS

Education & Certificates

Bachelor of Applied Science (BAS) in Computer Science — 2020

University of Pennsylvania

- Vice President, Artificial Intelligence Society; undergraduate thesis on differential privacy under Rajeev Alur, with distinction
- **DELE B2** (Instituto Cervantes) — official upper-intermediate Spanish language credential
- **AWS Certified AI Practitioner** — Verification ID: SM19EYTCJEV1QZ9X

Portfolio

Websites & Open Source:

- 1wg.ai (One World Gazette) — a Next.js news platform using RAG pipelines for AI-powered multilingual global news coverage.
- glottosphere.com — a universal translation and reference website leveraging FAISS similarity search and Cloudflare R2/Worker.
- vakyume — an LLM-assisted pipeline that turns textbook equations into verified Python and C++ solver libraries using SymPy.
- readme-rosetta — a Python CLI using Ollama, Rich, tqdm, and polib to translate Markdown READMEs across many languages.